Comparative Effectiveness of SGLT2 Inhibitor and GLP-1 Receptor Agonist on Kidney and Cardiovascular Outcomes by Kidney Failure Risk

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Key Points

- In veterans with type 2 diabetes and low kidney failure risk, sodium-glucose cotransporter 2 inhibitors (SGLT2is) were more kidney protective while glucagon-like peptide-1 receptor agonists (GLP-1 RAs) were more cardioprotective.
- For cardiovascular-kidney-metabolic outcomes, GLP-1 RAs were more protective at moderate kidney failure risk and SGLT2is were more protective at high kidney failure risk.
- The kidney failure risk equation might be a clinically useful tool to guide therapy in type 2 diabetes.

Abstract

Background Head-to-head comparisons of non-exendin glucagon-like peptide-1 receptor agonists (GLP-1 RAs) and sodium-glucose cotransporter 2 inhibitors (SGLT2is) on kidney failure or cardiovascular-kidney-metabolic (CKM) composite end points are lacking. Whether kidney failure risk modifies the comparative effectiveness of GLP-1 RA versus SGLT2i is clinically relevant.

Methods We defined a national veterans cohort with type 2 diabetes who initiated an SGLT2i, non-exendin GLP-1 RA, or insulin glargine between January 1, 2018, and December 31, 2021. After applying inverse probability of treatment weighting to balance baseline characteristics, outcomes were compared across new-user groups through March 31, 2023. Outcomes included kidney failure (stage 5 CKD or long-term KRT), major adverse cardiovascular events (MACEs: heart failure, myocardial infarction, or stroke), CKM composite (kidney failure or MACE), all-cause death, and composites of outcomes with death. We tested for effect modification by the kidney failure risk equation (KFRE) score on comparative pairwise drug effectiveness.

Results Out of 160,428 veterans, 53%, 14%, and 34% were new users of SGLT2i, GLP-1 RA, and insulin glargine, respectively. Relative to GLP-1 RA new-users, SGLT2i new-users had similar mortality risk, a trend toward lower kidney failure risk (hazard ratio [HR], 0.89; 95% confidence interval [CI], 0.74 to 1.06), but higher risk of MACE (HR, 1.14; 95% CI, 1.09 to 1.20) and CKM composite (HR, 1.13; 95% CI, 1.08 to 1.19). The effect of SGLT2i versus GLP-1 RA differed significantly between moderate-risk (2%–6%) and high-risk (≥6%) KFRE subgroups for all outcomes except all-cause death. In those with moderate-risk KFRE, GLP-1 RA seemed more protective for kidney failure, MACE, and CKM composite, while SGLT2i appeared more protective in those with high-risk KFRE.

Conclusions Compared with GLP-1 RA, SGLT2i had comparable risks of mortality, perhaps a lower risk of kidney failure, but modestly higher risk of cardiovascular events in the entire cohort. However, GLP-1 RA were more beneficial in those with moderate kidney failure risk and SGLT2i more beneficial in those with high kidney failure risk.

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Introduction

The cardiovascular, kidney, and metabolic systems are interdependent, and their dysfunction results in cardiovascular-kidney-metabolic (CKM) syndrome.[1] This syndrome is increasing in prevalence[2] and is associated with substantial morbidity and mortality.[1] Characterizing heterogeneity within CKM syndrome remains a research priority.[3]

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The American Diabetes Association recommends both sodium-glucose cotransporter 2 inhibitors (SGLT2is) and glucagon-like peptide-1 receptor agonists (GLP-1 RAs) for type 2 diabetes with CKD or cardiovascular disease,[4] but little evidence guides which to use first. The easy-to-use kidney failure risk equation (KFRE),[5–7] which predicts progression to long-term KRT, could help tailor therapy if comparative effectiveness of these agents varies across KFRE risk strata.

While SGLT2i consistently improve CKM outcomes,[8–15] GLP-1 RA exhibit more variability.[16–22] Exendin-4–based GLP-1 RA (exenatide[16] and lixisenatide[17]) show no cardiovascular benefits, while human GLP-1 analogs (liraglutide,[18] dulaglutide,[19] albiglutide,[20] and semaglutide[21,22]) demonstrated such benefits. Kidney-specific outcomes of GLP-1 RA are more heterogenous: Exenatide showed no benefit[23]; lixisenatide reduced new-onset macroalbuminuria only[24]; liraglutide reduced macroalbuminuria, but not doubling of serum creatinine or long-term KRT[25]; and dulaglutide reduced macroalbuminuria and ≥40% eGFR decline, but not kidney failure.[26]

The Evaluate Renal Function with Semaglutide Once Weekly (FLOW) trial was the first randomized, controlled trial specifically designed to examine the effects of a GLP-1 RA on hard kidney end points in type 2 diabetes and CKD.[21] Compared with placebo, semaglutide reduced the risk of (*1*) a composite of kidney failure, ≥50% reduction in eGFR, or death from kidney or cardiovascular causes and (*2*) a composite of kidney-specific components. However, unlike canagliflozin[8] or dapagliflozin,[9] semaglutide did not significantly reduce progression to stage 5 CKD (hazard ratio [HR], 0.80; 95% confidence interval [CI], 0.61 to 1.06) or long-term KRT (HR, 0.84; 95% CI, 0.63 to 1.12). Whether this reflects biological differences or trial design limitations is unclear.

Randomized head-to-head comparisons of GLP-1 RA and SGLT2i are lacking. Network meta-analyses[27–30] and target trial emulations[31–33] reported mixed cardiovascular effects. Three of these studies[29,30,33] compared kidney outcomes, but all predated FLOW[21] and included exendin GLP-1 RA agents.[16,17] To our knowledge, a focused comparison between non-exendin GLP-1 RA and SGLT2i for kidney failure outcomes has not been performed.

To address this gap, we conducted an active comparator, new-user design[34–38] study in a national cohort of veterans with type 2 diabetes who initiated one of these drug classes. The aims were to (*1*) compare the effectiveness of SGLT2i versus non-exendin GLP-1 RA on kidney failure, cardiovascular outcomes, and all-cause deaths and (*2*) evaluate whether baseline kidney failure risk (estimated by KFRE) modifies these comparative outcomes.

Methods

Data Sources

The Veterans Health Administration, the largest integrated health care system in the United States, served 9 million veterans in 2024.[39] We used the Veterans Affairs (VA) Informatics and Computing Infrastructure[40] to assess VA Corporate Data Warehouse[41] electronic health records that included diagnostic codes, imaging, laboratory values, inpatient and outpatient notes, prescription orders and fills, outpatient and inpatient diagnostic codes from community care,[42] and death data enriched with national death records.[43]

Study Design and Cohort Definition

To emulate a randomized, controlled trial, we used the active comparator, new-user design[34–37] aimed at minimizing confounding and bias.[38,44] Among 2,462,623 veterans with type 2 diabetes from January 1, 2018, to December 31, 2021 (Figure 1), there were 8,800,249 prescriptions for study drugs (Supplemental Table 1) representing 605,756 unique veterans. At least one study drug within each drug class of interest was offered by the Veterans Health Administration throughout the study period. The index date was the first date of study drug prescription fill during this period. After applying exclusion criteria (detailed in Supplemental Methods), the final cohort included 160,428 veterans with type 2 diabetes on metformin who were new-users of SGLT2i (N=84,625), non-exendin GLP-1 RA (N=21,788), or insulin glargine (N=54,015) (Figure 1).

Main Predictors

Treatment groups were defined by first-time initiation of SGLT2i, non-exendin GLP-1 RA, or insulin glargine added to metformin. Baseline kidney failure risk was estimated with the four-variable, 5-year KFRE[5–7] incorporating age, sex, eGFR by CKD Epidemiology Collaboration 2021 equation,[45] and urine albumin-creatinine ratio (UACR) with UACR values of 0 mg/g set to 0.001 mg/g. There were 70,589 (44%) with missing baseline UACR. In the absence of established thresholds,[6] KFRE categories were prespecified as low-risk (<2%), moderate-risk (2% to <6%), high-risk (≥6%), or missing.

Baseline Variables

Demographics were obtained from VA Corporate Data Warehouse[41] or Veterans Eligibility Trends and Statistics.[46] Baseline comorbidities, duration of type 2 diabetes, outpatient BP, height and weight, outpatient laboratory measurements, and medications were defined as described in Supplemental Methods and Supplemental Tables 2 and 3.

Follow-Up

Patients were followed until death, administrative censoring date of March 31, 2023, or last follow-up encounter (including community care data outside the VA system) before March 31, 2023. For each event of interest, censoring date was defined as time to the occurrence of the event or death, administrative censoring date, or last follow-up encounter before March, 31, 2023. Supplemental Table 4 summarizes last follow-up dates available for each outcome.

Long-Term Medication Use

New-users of SGLT2i and GLP-1 RA who were eligible for receiving the medication (no death, medication discontinuation, loss to follow-up, or administrative censoring events) were followed over 90-day intervals for discontinuation (prescription fills covering 0 out of 90 days for

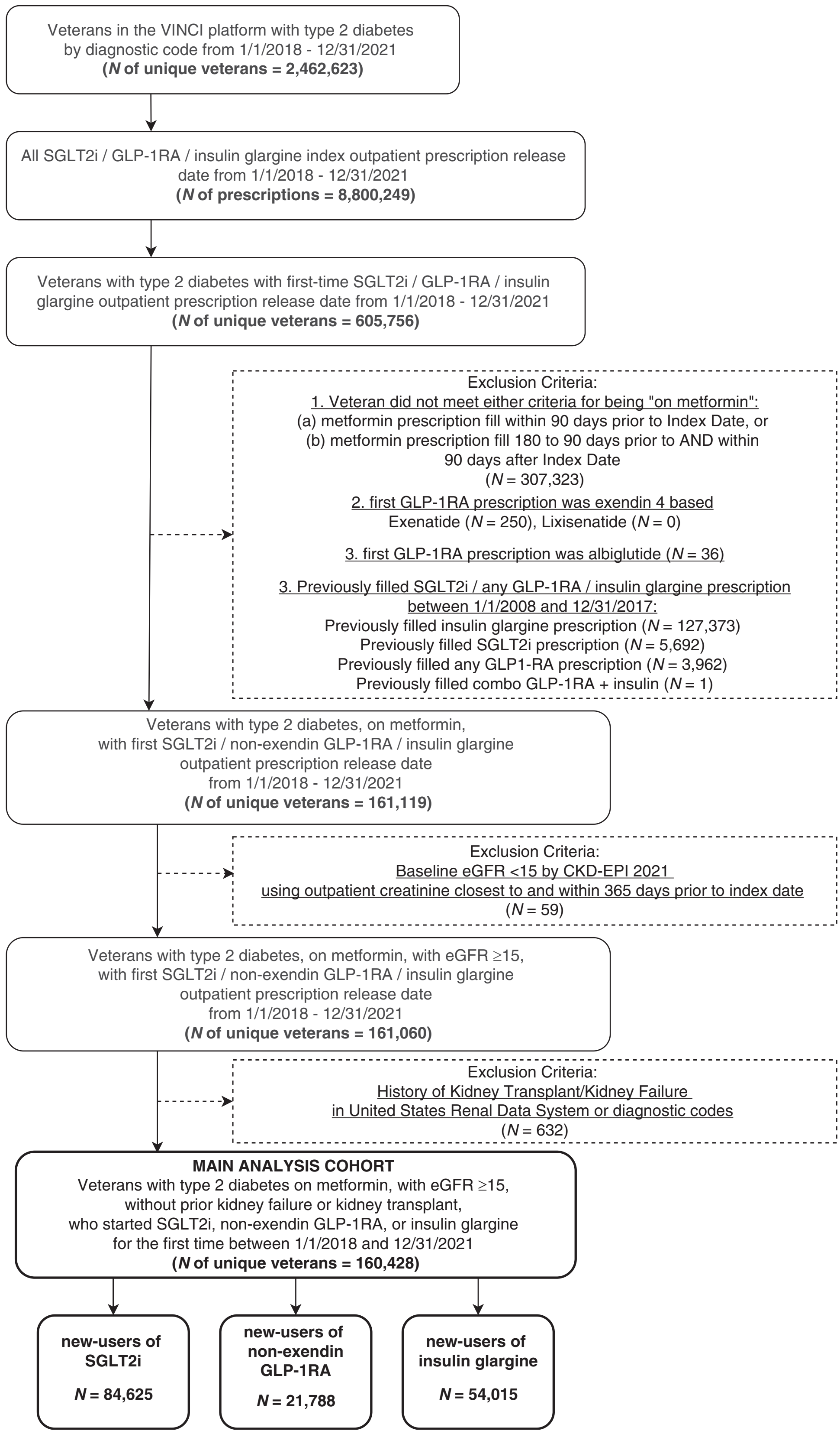

Figure 1. Study cohort flow chart. CKD-EPI, CKD Epidemiology Collaboration; GLP-1 RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium-glucose cotransporter 2 inhibitor; VINCI, VA Informatics and Computing Infrastructure.

two subsequent 90-day intervals) and class-switching (SGLT2i discontinuers who had prescription fill for GLP-1 RA in the same 90-day period as discontinuation and *vice versa*).

Outcome Variables

We defined kidney failure as sustained stage 5 CKD or need for long-term KRT. We defined sustained stage 5 CKD as two consecutive outpatient or inpatient CKD Epidemiology Collaboration 2021[45] eGFR <15 ml/min per 1.73 m^2 at least 60 days apart, with no interim values ≥15 ml/min per 1.73 m^2. Individuals missing a second creatinine because of loss of follow-up, death, or long-term KRT were classified as meeting stage 5 CKD criteria. We defined need for long-term KRT as long-term dialysis dependence or kidney transplantation, obtained through December 31, 2021, with United States Renal Data System[47] linkage through the VA Information Resource Center[48] and through March 31, 2023, with diagnostic codes for long-term KRT or transplant in VA Corporate Data Warehouse (Supplemental Table 5).

We defined major adverse cardiovascular events (MACEs) as the first hospitalization or emergency room occurrence of myocardial infarction, stroke, transient ischemic attack, or heart failure identified *via International Statistical Classification of Diseases, Tenth Revision,*[49] codes from VA and fee-for-service non-VA community care data (Supplemental Table 5). The CKM composite was defined as the first occurrence of kidney failure or MACE. All-cause death was obtained from VA Vital Status database,[43] which integrates data from Centers for Medicaid and Medicare services, Social Security Administration, and the VA Austin Information Technology Center. We also examined composite outcomes of all-cause death with each of the above outcomes.

Statistical Methods

Data Preparation

Clinical measures data outside prespecified plausible ranges were set as missing. Missingness is summarized in Supplemental Table 6. We handled missing data with K-nearest neighbors–based imputation[50,51] (using five neighbors) because this approach balances accuracy and computational efficiency for handling large-scale electronic health record data.[50,51] However, for missing UACR data, we used a missing category because of its high frequency.

Weighting

Inverse probability of treatment weighting was used to achieve balance across treatment groups for all baseline variables defined in Supplemental Table 7. Details and weighting diagnostics are provided in Supplemental Methods.

Baseline Characteristics

Characteristics before and after weighting were summarized by treatment group.

Comparative Effectiveness of SGLT2i versus GLP-1 RA on CKM Outcomes in the Entire Cohort

We generated weighted cumulative incidence curves for kidney failure, MACE, and CKM composite with death as competing risk. Under this competing risk framework, 3-year absolute risk and absolute risk differences were derived from these functions with 1000 bootstraps to generate 95% CIs. For time-to-event analyses, we used weighted cause-specific Cox proportional hazards models because they are well-suited for etiologic and mechanistic questions because they estimate the instantaneous kidney and/or cardiovascular risks among individuals still at risk for therapy, without inducing spurious associations with competing events such as death.[52,53] We tested proportional hazards assumptions using time-by-treatment interactions, dichotomizing follow-up at 2 years to assess whether any change in trend around this time point was significant. No significant proportional hazards violations were identified (Supplemental Table 8).

Comparative Effectiveness of SGLT2i versus GLP-1 RA on CKM Outcomes within KFRE Subgroups

We repeated absolute risk estimation and weighted Cox regressions within each KFRE subgroup and tested whether the comparative effectiveness of SGLT2i versus GLP-1 RA was modified in moderate-risk versus high-risk KFRE subgroups with a Wald-type comparison for absolute risk differences and Wald test comparison of regression coefficients in the relative scale.

Sensitivity Analyses

We performed alternative analyses by baseline eGFR and albuminuria (Supplemental Methods).

Analyses were performed using STATA, version 18 (StataCorp, LLC, College Station, TX), and R version 4.3.1 (R Foundation for Statistical Computing, Vienna, Austria). Statistical significance was defined as two-sided P value < 0.05. Given the retrospective nature of this study and the multiple comparisons tested, all P values reported are nominal. This study was approved by the University of Utah Institutional Review Board and the Salt Lake City VA Research and Development Committee.

Results

There were 84,625 (53%) new-users of SGLT2i, 21,788 (14%) new-users of non-exendin GLP-1 RA, and 54,015 (34%) new-users of insulin glargine. New-users of SGLT2i were slightly older and more likely to have a history of cardiovascular disease. New-users of GLP-1 RA were more likely to be female and had higher body mass index. New-users of insulin glargine were more likely non-White, had lower educational status and income, were more likely to have chronic lung disease, cancer, substance use disorders, and higher hemoglobin A1c (Table 1). There was good overlap of treatment propensities by treatment group (Supplemental Figure 1). Inverse probability of treatment weighting effectively balanced baseline characteristics across study drug groups (Supplemental Figures 2 and 3).

Overall mean baseline eGFR was 81±19 ml/min per 1.73 m^2. There were 70,589 (44%) individuals with missing UACR. Median UACR was 21 (interquartile range [IQR], 9–65) mg/g. Baseline characteristics by UACR groups, including those with missing UACR as a separate category, are summarized in Supplemental Table 9. Weighted baseline characteristics across SGLT2i and GLP-1 RA

Table 1. Baseline characteristics by study drug groups (*N*=160,428), unweighted and after inverse probability of treatment weighting

Characteristic	Unweighted			Inverse Probability of Treatment Weighted		
	SGLT2i (*N*=84,625[a,b])	GLP-1 RA (*N*=21,788[a,b])	Insulin Glargine (*N*=54,015[a,b])	SGLT2i[b]	GLP-1 RA[b]	Insulin Glargine[b]
Demographics						
Age, yr	66±10	64±11	64±11	65±11	65±11	65±11
Female, *n* (%)	3577 (4)	1967 (9)	3043 (6)	6%	6%	5%
Race, *n* (%)						
Black or African American	14,404 (17)	3596 (17)	11,763 (22)	19%	18%	19%
Other/mixed/unknown	4649 (6)	1222 (6)	2563 (5)	5%	6%	5%
White	65,572 (78)	16,970 (78)	39,689 (74)	76%	77%	76%
Hispanic/Latino ethnicity, *n* (%)	9756 (12)	2222 (10)	5866 (11)	11%	11%	12%
Completed high school, *n* (%)	33,964 (40)	8770 (40)	21,256 (39)	40%	41%	40%
Household income <$50,000/yr, *n* (%)	39,647 (47)	9967 (46)	27,610 (51)	48%	48%	49%
Clinical characteristics						
Duration of type 2 diabetes, yr	9±6	8±6	9±6	9±6	9±6	9±6
Coronary artery disease, *n* (%)	36,653 (43)	7517 (35)	19,343 (36)	40%	39%	40%
Heart failure, *n* (%)	14,781 (18)	2809 (13)	7982 (15)	16%	16%	16%
Cardiomyopathy, *n* (%)	16,353 (19)	3073 (14)	8454 (16)	17%	17%	18%
Cerebrovascular disease, *n* (%)	14,713 (17)	3232 (15)	9769 (18)	18%	16%	18%
Peripheral vascular disease, *n* (%)	20,224 (24)	4533 (21)	11,826 (22)	23%	22%	23%
Hypertension, *n* (%)	77,767 (92	19,777 (91)	48,343 (90)	91%	91%	91%
Atrial fibrillation/flutter, *n* (%)	11,959 (14)	2519 (12)	6740 (13)	13%	13%	13%
Chronic lung, *n* (%) disease/sleep apnea	51,417 (61)	14,137 (65)	31,454 (58)	60%	60%	60%
Cancer, *n* (%)	12,807 (15)	3068 (14)	9306 (17)	16%	15%	16%
HIV-AIDS, *n* (%)	424 (0.5)	123 (0.6)	365 (0.7)	0.6%	0.9%	0.5%
Traumatic brain injury, *n* (%)	2169 (3)	570 (3)	1615 (3)	3%	3%	3%
Dementia, *n* (%)	1976 (2)	535 (3)	2307 (4)	3%	3%	3%
Post-traumatic stress disorder, *n* (%)	23,566 (28)	6497 (30)	15,863 (29)	29%	28%	29%
Depression, *n* (%)	37,381 (44)	10,460 (48)	26,542 (49)	47%	46%	45%
Tobacco use disorder, *n* (%)	26,735 (32)	6154 (28)	19,202 (36)	33%	32%	32%
Alcohol use disorder, *n* (%)	12,196 (14)	2774 (13)	9598 (18)	16%	14%	15%
Diabetic retinopathy, *n* (%)	21,523 (25)	5476 (25)	14,358 (27)	26%	26%	27%
Diabetic neuropathy, *n* (%)	28,015 (33)	8009 (37)	19,113 (35)	35%	35%	35%
Physical measurements						
Systolic BP, mm Hg	137±17	135±16	135±16	136±16	136±17	136±17
Diastolic BP, mm Hg	78±9	78±9	78±9	78±9	78±9	78±9
BP day count[c]	5±7	6±7	7±11	6±12	6±8	6±8
BMI, kg/m^2	33.2±6.0	35.5±6.8	32.5±6.4	33.3±6.2	33.6±6.1	33.3±6.6
Medications, *n* (%)						
DPP-4i	25,302 (30)	5963 (27)	12,108 (22)	27%	28%	27%
Sulfonylureas	45,060 (53)	11,741 (54)	27,143 (50)	52%	54%	52%
Thiazolidinediones	6798 (8)	2185 (10)	3630 (7)	8%	9%	8%
Other insulins	8.655 (10)	4728 (22)	18,661 (35)	21%	21%	21%
Diuretics	26,938 (32)	7091 (33)	16,300 (30)	32%	31%	32%

Table 1. Continued

Characteristic	Unweighted			Inverse Probability of Treatment Weighted		
	SGLT2i (N=84,625[a,b])	GLP-1 RA (N=21,788[a,b])	Insulin Glargine (N=54,015[a,b])	SGLT2i[b]	GLP-1 RA[b]	Insulin Glargine[b]
ACEi, ARB, or angiotensin receptor/neprilysin inhibitors	59,763 (71)	15,229 (70)	34,882 (65)	69%	69%	69%
β blocker	36,730 (43)	8327 (38)	20,155 (37)	41%	41%	40%
Calcium channel blocker	24,242 (29)	6022 (28)	14,883 (28)	28%	29%	28%
Statin	69,680 (82)	17,559 (81)	40,355 (75)	80%	81%	80%
Laboratory measurements						
Hemoglobin A1c, %	8.6±1.5	8.8±1.7	9.9±2.3	9.1±1.9	9.0±1.8	9.0±2.0
Hemoglobin A1c, %, *n* (%)						
<8	31,463 (37)	6970 (32)	11,320 (21)	30%	29%	34%
8 to <9	25,417 (30)	5763 (27)	9741 (18)	27%	25%	22%
9 to <10	15,399 (18)	4269 (20)	9583 (18)	19%	20%	17%
≥10	12,346 (15)	4786 (22)	23,371 (43)	24%	26%	28%
eGFR, ml/min per 1.73 m^2	80±18	81±20	81±21	80±19	80±20	80±20
eGFR, ml/min per 1.73 m^2, *n* (%)						
15 to <30	69 (0.1)	88 (0.4)	316 (0.6)	0.1%	0.4%	0.5%
30 to <45	2248 (3)	1004 (5)	2345 (4)	3%	5%	4%
45 to <60	10,999 (13)	2737 (13)	7252 (13)	14%	13%	13%
≥60	71,309 (84)	17,959 (82)	44,102 (82)	84%	82%	82%
UACR, mg/g, *n* (%)						
<30	29,273 (60)	7560 (61)	15,942 (56)	58%	58%	59%
30 to <100	10,927 (22)	2662 (22)	6843 (24)	23%	23%	23%
100 to <300	5180 (11)	1268 (10)	3317 (12)	11%	11%	11%
≥300	3608 (7)	889 (7)	2370 (8)	8%	8%	8%
UACR missing, *n* (%)	35,637 (4)	9409 (43)	25,543 (47)	44%	44%	45%
5-yr four-variable KFRE, *n* (%)						
<2% (low)	46,851 (96)	11,614 (94)	26,541 (93)	95%	93%	95%
2% to <6% (moderate)	1525 (3)	511 (4)	1217 (4)	4%	5%	4%
≥6% (high)	612 (1)	254 (2)	714 (3)	2%	2%	2%
KFRE missing, *n* (%)	35,637 (42)	9409 (43)	25,543 (47)	44%	44%	45%

ACEi, angiotensin-converting-enzyme inhibitor; ARB, angiotensin receptor blocker; BMI, body mass index; DPP-4i, dipeptidyl peptidase 4 inhibitor; GLP-1 RA, glucagon-like peptide-1 receptor agonist; KFRE, kidney failure risk equation; SGLT2i, sodium-glucose cotransporter 2 inhibitor; UACR, urine albumin-creatinine ratio.

[a]Please note that the distribution of study drug groups is 84,625 (52.7%) new-users of sodium-glucose cotransporter 2 inhibitor, 21,788 (13.6%) new-users of non-exendin glucagon-like peptide-1 receptor agonist, and 54,015 (33.7%) new-users of insulin glargine, but that after rounding to 53%, 14%, and 34%, respectively, the total is 101%.

[b]All continuous covariates are summarized as mean±SD; all categorical covariates are summarized as *n* (%).

[c]Number of days with BP values taken, regardless of more than one value on a given day.

initiators were balanced in the missing UACR subgroup (Supplemental Figure 4).

We could not calculate KFRE scores in 70,589 (44%) veterans due to missing UACR. There were 85,006 (95%) veterans with low-risk, 3253 (4%) with moderate-risk, and 1580 (2%) with high-risk KFRE. The median KFRE scores in those with low-, moderate-, and high-risk KFRE were 0.03% (IQR, 0.01%–0.15%), 3% (IQR, 3%–4%), and 11% (IQR, 8%–18%), respectively. The distribution of KFRE risk subgroups was well-balanced across the study drug groups after inverse probability of treatment weighting (Table 1 and Supplemental Figure 2).

There were 1633 kidney failure composite events over 442,193 person-years of follow-up (event rate 3.7/1000 person-years), 20,557 MACE events over 411,975 person-years of follow-up (event rate 49.9/1000 person-years), 21,272 CKM composite events over 411,169 person-years of follow-up (event rate 51.7/1000 person-years), and 15,099 deaths over 443,626 patient-years of follow-up (event rate 34.0/1000 person-years). As shown in Supplemental Table 4, <5% of patients were lost to follow-up 6 months or more before the administrative censoring date. Unweighted and weighted event rates for outcomes within the drug classes and KFRE subgroups are presented in Supplemental Table 10. Of note, event rates in the KFRE missing subgroup were closest to the low-risk KFRE subgroup. Furthermore, KFRE stratified the study population by kidney failure risk, with observed kidney failure rates of 2.4, 12.7, and 41.9 per 1000 patient-years in low-, moderate-, and high-risk groups, respectively.

Comparative Effectiveness of SGLT2i versus GLP-1 RA on CKM Outcomes in the Entire Cohort

Compared with GLP-1 RA, SGLT2i seemed to have similar incidence of kidney failure, all-cause death, and kidney failure/all-cause death composite but higher incidence of MACE, MACE/all-cause death composite, CKM composite, and CKM composite/all-cause death composite (Figure 2 and Supplemental Figure 5).

Absolute risk differences in comparative effectiveness of SGLT2i and GLP-1 RA mirrored the relative risk differences as summarized in Figure 3. Compared with those initiating GLP-1 RA, the 3-year absolute risk difference in SGLT2i initiators for kidney failure was nonsignificant (−0.13%; 95% CI, −0.34% to 0.07%), but was higher for MACE (1.31%; 95% CI, 0.71% to 1.92%) and CKM composite (1.20%; 95% CI, 0.54% to 1.85%) (Figure 3A). While absolute risk differences for all-cause death and kidney failure/all-cause death were nonsignificant, SGLT2i had significantly higher absolute risk compared with GLP-1 RA for MACE/all-cause death and CKM composite/all-cause death (Figure 3A).

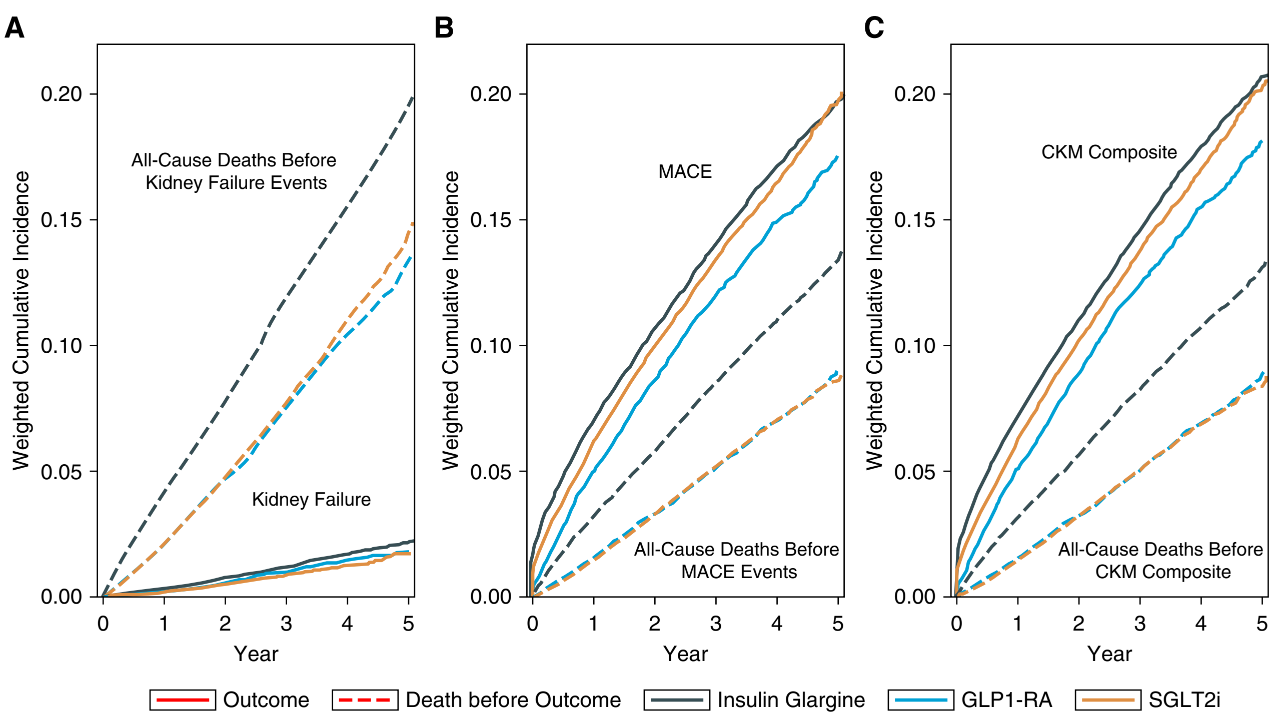

Figure 2. Cumulative incidence of outcomes among new-users of SGLT2i (*n*=84,625), GLP-1 RA (*N*=21,788), and insulin glargine (*N*=54,015). Cumulative incidence curves are inverse probability of treatment weighted and generated under a competing risk framework. In each plot, the solid line represents the outcome, and the dashed line represents all-cause death before that outcome. The dark gray line represents new-users of insulin glargine, the blue line represents new-users of GLP-1 RA, and the yellow line represents new-users of SGLT2i. (A) Demonstrates the cumulative incidence of the outcome of kidney failure (sustained stage 5 CKD stage 5 or long-term KRT) in solid lines and all-cause deaths before kidney failure events in dashed lines. (B) Shows the cumulative incidence of MACE in solid lines and all-cause deaths before MACE events in dashed lines. (C) Shows the cumulative incidence of the CKM composite outcome (kidney failure or MACE) in solid lines and all-cause deaths before CKM events in dashed lines. CKM, cardiovascular-kidney-metabolic; MACE, major adverse cardiovascular events.

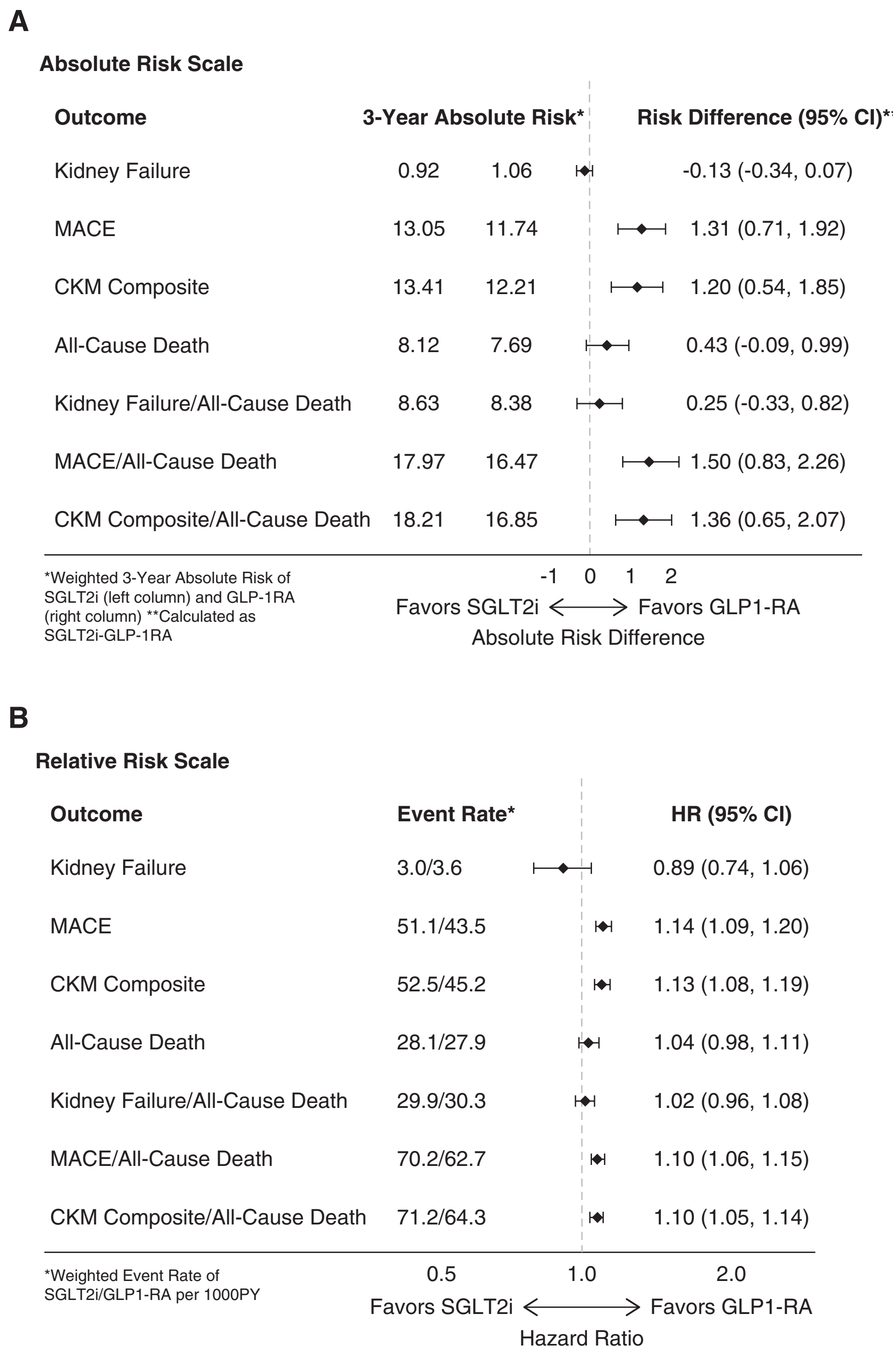

Figure 3. Comparative effectiveness of SGLT2i (*n*=84,625) versus non-exendin GLP-1 RA (*n*=21,788) in the entire cohort. This figure shows a forest plot of inverse probability of treatment weighted absolute risk at 3 years and absolute risk difference (A) and inverse probability of treatment weighted Cox regression results (B) demonstrating the comparative effectiveness of SGLT2i versus GLP-1 RA for all outcomes, with GLP-1 RA as reference. Kidney failure represents a composite of sustained stage 5 CKD or long-term KRT events. MACE represents a composite of MI, heart failure, or stroke events. CKM composite represents a composite of kidney failure or MACE events. Each of these outcomes is also depicted as a composite with all-cause death. Absolute risk difference represents the 3-year absolute risk of the SGLT2i group minus the GLP-1 RA group. Event rates depict rates in SGLT2i group/GLP-1 RA group, per 1000 person-years. CI, confidence interval; MI, myocardial infarction.

Similarly, in weighted Cox regression models (Figure 3B), those initiating SGLT2i had a modest and nonsignificantly lower hazard of kidney failure (HR, 0.89; 95% CI, 0.74 to 1.06) and a modestly higher hazard of MACE (HR, 1.14; 95% CI, 1.09 to 1.20) and CKM composite events (HR, 1.13; 95% CI, 1.08 to 1.19) compared with those initiating GLP-1 RA. While the hazard of all-cause death and kidney failure/all-cause death were similar, hazards for MACE/all-cause death (HR, 1.10; 95% CI, 1.06 to 1.15) and CKM composite/all-cause

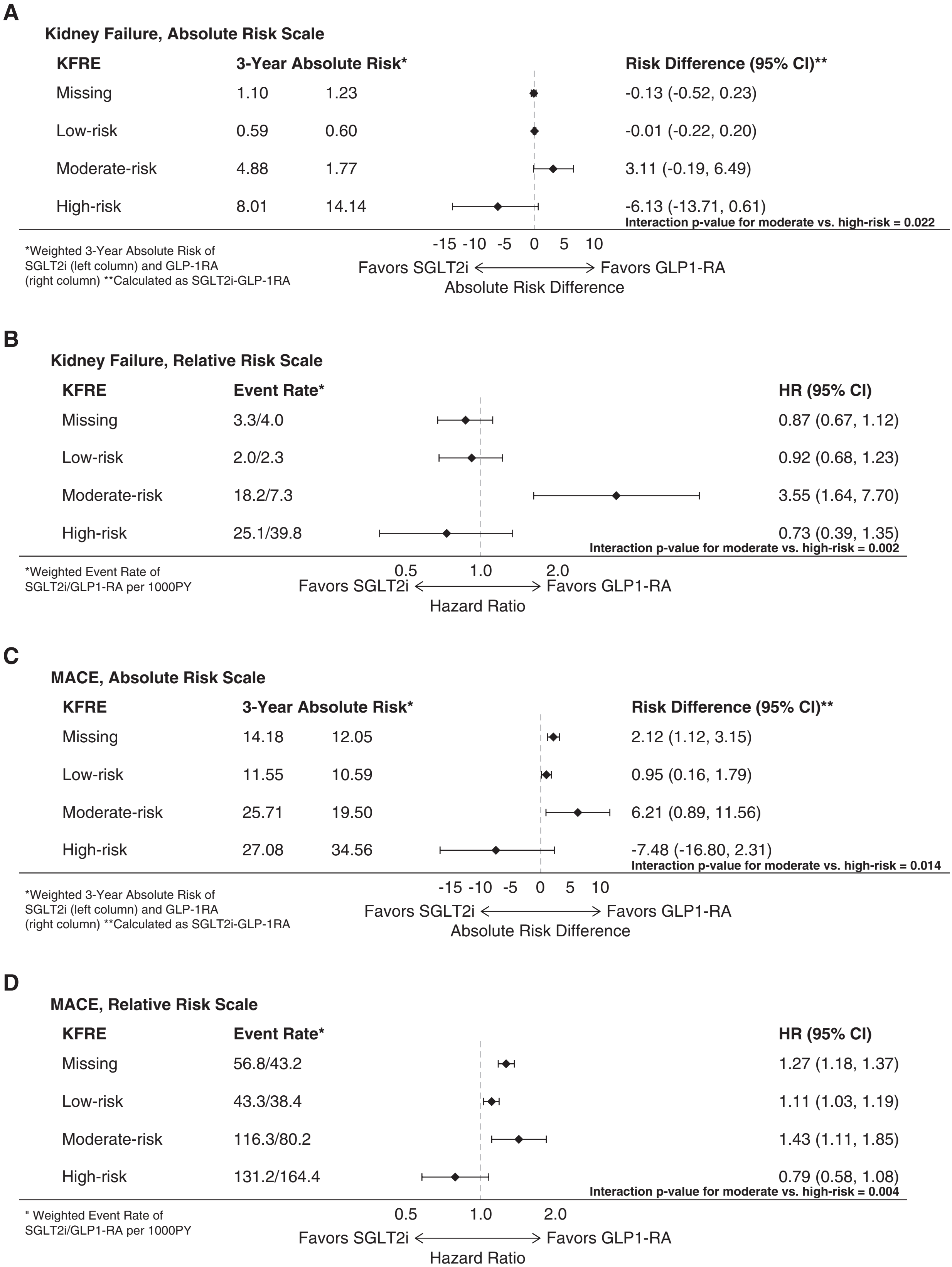

Figure 4. Comparative effectiveness of SGLT2i (*n*=84,625) versus non-exendin GLP-1 RA (*n*=21,788) across baseline KFRE subgroups. This figure shows forest plots of inverse probability of treatment weighted absolute risk at 3 years and absolute risk difference (A, C, E, and G) and inverse probability of treatment weighted Cox regression results (B, D, F, and H) demonstrating comparative effectiveness of SGLT2i versus GLP-1 RA within baseline KFRE subgroups, with GLP-1 RA as reference. Absolute risk difference represents the 3-year

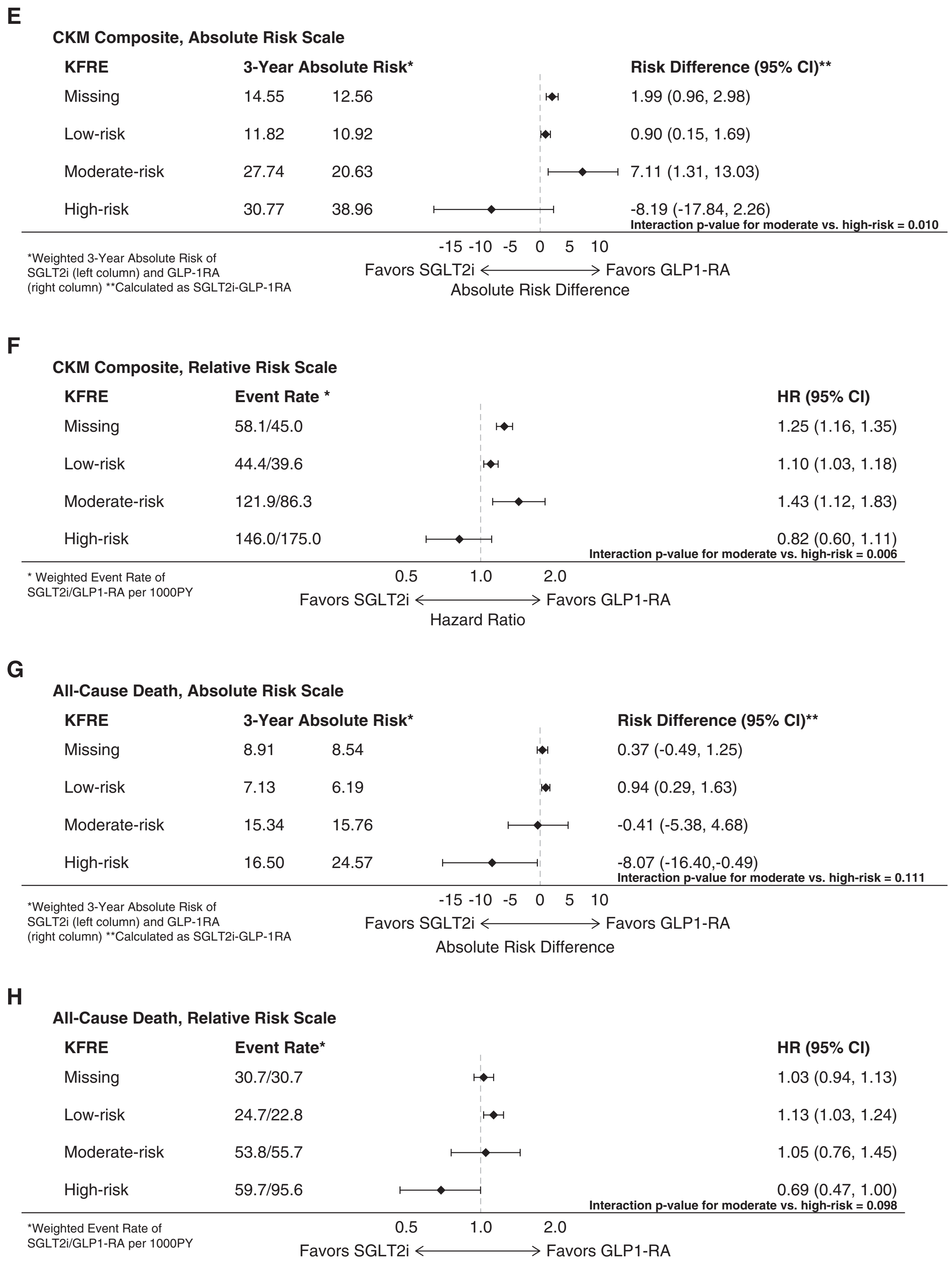

Figure 4. *Continued.* absolute risk of the SGLT2i group minus the GLP-1 RA group. Event rates are depicted as weighted rates in SGLT2i/GLP-1 RA group, per 1000 person-years. Interaction *P* values between the moderate-risk and high-risk KFRE subgroups are displayed in the lower right corner of each panel. (A and B) Show the kidney failure outcome (composite of sustained stage 5 CKD or

Figure 4. *Continued.* long-term KRT events). (C and D) Show MACE, a composite of MI, heart failure, or stroke events. (E and F) Show the CKM composite outcome of kidney failure or MACE events. (G and H) Show all-cause death events. KFRE, kidney failure risk equation.

death (HR, 1.10; 95% CI, 1.05 to 1.14) remained modestly higher in SGLT2i initiators compared with GLP-1 RA initiators.

As summarized in Supplemental Table 11, compared with SGLT2i or GLP-1 RA, insulin glargine had higher hazards of all outcomes.

Comparative Effectiveness of SGLT2i versus GLP-1 RA on CKM Outcomes within KFRE Subgroups

Baseline variables were largely well-balanced across study drug pairs after weighting within KFRE subgroups (Supplemental Figure 6). In general, the absolute risk differences and hazard ratios for the comparative effectiveness of SGLT2i versus GLP-1 RA for each of the outcomes in those with missing KFRE scores were similar to the low-risk KFRE subgroup and mirrored trends seen in the overall cohort (Figure 4, lines 1 and 2).

In the moderate-risk KFRE subgroup, compared with GLP-1 RA initiators, SGLT2i initiators had higher absolute and relative risks of kidney failure, MACE, and CKM composite, but similar risk of all-cause death (Figure 4, line 3). By contrast, SGLT2i seemed more beneficial in the high-risk KFRE subgroup with nonsignificant, but lower, risk of kidney failure, MACE, and CKM composite and perhaps a significantly lower risk of all-cause death (Figure 4, line 4). Interaction *P* values for comparisons of SGLT2i versus GLP-1 RA in moderate-risk and high-risk KFRE subgroups were significant for kidney failure, MACE, and CKM composite, but not for all-cause death in both absolute and relative scales as presented in Figure 4.

Results were similar when considering outcomes as composites with all-cause death (Supplemental Figure 7). Corresponding comparative effectiveness of insulin glargine versus SGLT2i or GLP-1 RA by KFRE subgroups are presented in Supplemental Table 12.

Long-Term Medication Use

Among GLP-1 RA new-users, 85%, 76%, and 70% continued on a GLP-1 RA at months 12, 24, and 36, respectively (Supplemental Figure 8). The corresponding numbers for SGLT2i were slightly lower at 82%, 72%, and 66%, respectively. Among those who discontinued GLP-1 RA or SGLT2i, higher proportions switched to the other drug class later in the study period, reflecting greater adoption of these agents in clinical practice (Supplemental Table 13).

Sensitivity Analyses

Trends by eGFR or UACR subgroup largely mirrored those by KFRE subgroup (Supplemental Tables 14 and 15).

Discussion

In this large, real-world study of veterans with type 2 diabetes, compared with GLP-1 RA, SGLT2i initiation had comparable risk of mortality, perhaps a lower risk of kidney failure, but modestly higher risk of cardiovascular events in the entire cohort. However, the risk of kidney failure seems to modify the comparative effectiveness: GLP-1 RA might be more beneficial in those at moderate risk of kidney failure and SGLT2i more beneficial in those at high risk of kidney failure.

As CKM syndrome reflects the interdependence of kidney, cardiovascular, and metabolic disease,[1] CKM composite is a clinically useful construct. However, MACE occurred far more frequently (13.5-fold higher) than kidney failure in this study and thus the CKM composite was dominated by MACE events. As a result, an intervention may seem protective against CKM events, even if it has no kidney benefit. Hence, it is important to examine the comparative effectiveness of interventions on kidney outcomes alone. Kidney failure is the definitive kidney outcome and the ultimate kidney hard end point, as opposed to surrogate end points, such as albuminuria, slope of eGFR, or percent decline in eGFR. The KFRE was developed to predict kidney failure.

In pivotal trials, SGLT2i canagliflozin (HR, 0.68; 95% CI, 0.54 to 0.86)[8] and dapagliflozin (HR, 0.64; 95% CI, 0.50 to 0.82)[9] significantly reduced the hazard of kidney failure compared with placebo. The Study of Heart and Kidney Protection with Empagliflozin (EMPA-KIDNEY)[10] study reported the effects of empagliflozin on a kidney progression composite that included kidney failure, sustained 40% decline in eGFR or kidney death, but not the effects on kidney failure *per se*. Similarly, as reviewed in the introduction, the GLP-1 RA semaglutide significantly lowered the risk of a primary composite outcome including kidney- or cardiovascular-related death, but not of stage 5 CKD nor long-term KRT.[21]

In the absence of head-to-head trials comparing SGLT2i versus GLP-1 RA on kidney outcomes, network meta-analyses of trials of SGLT2i or GLP-1 RA against a common comparator provide indirect evidence of their comparative effectiveness. Two network meta-analyses examined kidney outcomes and both found that SGLT2i were superior to GLP-1 RA in prevention of composite kidney end points.[29,30] These studies included exendin GLP-1 RA that have not demonstrated kidney-[23,24] or cardiovascular-protective effects[16,17] and were conducted before the FLOW results[21] were published.

Emulated trials using observational data may inform comparative effectiveness of SGLT2i versus GLP-1 RA on kidney failure. Edmonston *et al.*[33] compared SGLT2i (n=35,004) versus GLP-1 RA (n=47,268) using electronic health record data from 20 US health systems. The composite kidney outcome (≥40% eGFR decline, long-term KRT, or all-cause death) did not differ between treatments (HR, 0.91; 95% CI, 0.81 to 1.02), but SGLT2i had lower risk of ≥40% eGFR decline (HR, 0.77; 95% CI, 0.65 to 0.91). Key limitations were inclusion of exendin GLP-1 RA, short follow-up (1.4 years), and long-term KRT assessment limited to the data within the health care systems.

To our knowledge, our study was the first to compare non-exendin GLP-1 RA (semaglutide, liraglutide, and dulaglutide) with SGLT2i for kidney failure using linkage to the United States Renal Data System[47] and the first to examine whether KFRE[5,7] modified their comparative effectiveness. The results of the current study suggest that, compared with GLP-1 RA, SGLT2i had similar mortality risk, nonsignificantly lower risk of kidney failure (HR, 0.89; 95% CI, 0.74 to 1.06), but higher risk of MACE (HR, 1.14; 95% CI, 1.09 to 1.20) and CKM composite outcome (HR, 1.13; 95% CI, 1.08 to 1.19). Although nonsignificant, the 95% CI (26% lower risk to 6% higher risk) for kidney failure predominantly favored SGLT2i and the possibility that SGLT2i may be more beneficial for kidney failure should not be ruled out. Indeed, the risk for kidney failure modified the comparative effectiveness of SGLT2i versus GLP-1 RA because SGLT2i seemed more beneficial for kidney and cardiovascular outcomes in those with high risk of kidney failure (KFRE ≥6%). Conversely, GLP-1 RA seemed more beneficial in those at moderate risk of kidney failure (KFRE 2%–6%).

Although KFRE was validated for kidney failure prediction only in individuals with eGFR <60 ml/min per 1.73 m^2,[5,54] we used it to stratify kidney failure risk rather than to predict kidney failure events. The observed kidney failure rates of 2.4, 12.7, and 41.9 per 1000 patient-years in low-, moderate-, and high-risk groups, respectively, demonstrated that KFRE effectively identified a high–kidney failure risk group. Nonetheless, wider CIs for SGLT2i versus GLP-1 RA comparisons in the high–KFRE risk group and narrower CIs in the low–KFRE risk group support greater certainty of the results in type 2 diabetes patients at low kidney failure risk, who represent the majority of our cohort.

Current clinical practice guidelines for CKD management in the setting of type 2 diabetes from the VA[55] and International Society of Nephrology[54] recommend SGLT2i before GLP-1 RA. Resulting restrictions on access to GLP-1 RA deserve evaluation. Furthermore, our results support policy changes to enhance access to SGLT2i or GLP-1 RA, both of which were superior to insulin glargine for kidney, cardiovascular, and mortality protection.

Our study incorporated several strengths, including an active comparator, new-user design, inverse probability of treatment weighting, and causal inference methods applied to rich electronic health records. These design features minimized confounding and reduced channeling, immortal time, survivor, and prevalent user biases. The VA setting limited financially driven prescription patterns. Overall, the study emulated a target trial of GLP-1 RA versus SGLT2i, enhancing internal validity. Concordance of absolute and relative scales further supports the robustness of our findings.

Our study also had several limitations. The population was primarily older White males, which limits broader applicability. A large share of missing UACR values resulted in many missing KFRE scores. However, the missing KFRE subgroup had a risk of kidney failure and comparative effectiveness of SGLT2i versus GLP-1 RA that was closest to those observed in the low-risk KFRE subgroup. Missing UACR values similarly affect CKD Prognosis Consortium[56] and Klinrisk[57] scores that also require UACR. The study does not clarify biological mechanisms for the interactions of KFRE with comparative effectiveness of SGLT2i and GLP-1 RA. Future per-protocol perspective analyses would help capture effects of class switching and other effects of adherence to the study protocol. However, we observed similar discontinuation rates and class switching with SGLT2i and GLP-1 RA. Finally, as with any retrospective study, unmeasured confounding remains a possible limitation, especially for hazard ratios closer to one.

In summary, compared with non-exendin GLP-1 RA, SGLT2i had comparable risk of mortality, perhaps a lower risk of kidney failure, but modestly higher risk of cardiovascular events in the entire cohort of veterans with type 2 diabetes. Similar patterns were seen in those at lower risk of kidney failure, who represent the majority of our cohort. However, GLP-1 RA appeared more effective against kidney failure and CKM outcomes in those with moderate risk of kidney failure while SGLT2i seemed more effective in those with high risk of kidney failure.

Disclosures

Disclosure forms, as provided by each author, are available with the online version of the article at http://links.lww.com/JSN/F653.

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Visualization: Robert E. Boucher, Sydney E. Hartsell, Ravinder Singh, Guo Wei.
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Declarative Statements

This study includes clinical experimentation and received Institutional Review Board or Ethics Committee approval. The need to obtain informed patient consent was waived. This study includes clinical experimentation and complies with the Declaration of Helsinki.

Data Availability Statements

Data belong to a third party, and authors are not authorized to share the data. Third Party: US Department of VA. Reason for Restriction: US Department of VA Policy.

Supplemental Material

This article contains the following supplemental material online at http://links.lww.com/JSN/F654.

Supplemental Methods

Supplemental Figure 1. Inverse probability of treatment weighting overlap diagnostics showing the distribution of treatment propensities by treatment group, before and after truncation.

Supplemental Figure 2. Balance plots for overall study drug comparisons (A) as well as pairwise study drug comparisons (B: GLP-1 RA versus SGLT2i. C: insulin glargine versus GLP-1 RA. D: insulin glargine versus SGLT2i).

Supplemental Figure 3. Standardized mean differences for the comparison between SGLT2i and GLP-1 RA groups across tertiles of the probability of receiving SGLT2i.

Supplemental Figure 4. Balance plot for pairwise study drug comparison GLP-1 RA versus SGLT2i within those missing UACR.

Supplemental Figure 5. Cumulative incidence of death-composite outcomes among new-users of SGLT2i (n=84,625), GLP-1 RA (N=21,788), and insulin glargine (N=54,015).

Supplemental Figure 6. Balance plots for GLP-1 RA versus SGLT2i comparisons within KFRE subgroups of missing (A), low-risk (KFRE <2%) (B), moderate-risk (KFRE 2% to <6%) (C), and high-risk (KFRE ≥6%) (D).

Supplemental Figure 7. Comparative effectiveness of SGLT2i (n=84,625) versus non-exendin GLP-1 RA (n=21,788) across baseline KFRE subgroups for death-composite outcomes.

Supplemental Figure 8. Index therapy continuation among new-users of SGLT2i and GLP-1 RA over 3 years.

Supplemental Table 1. Medications included in study drug groups.

Supplemental Table 2. Diagnostic codes used to define comorbidities for inclusion, exclusion, and baseline variables.

Supplemental Table 3. Baseline medications by category.

Supplemental Table 4. Last follow-up dates by outcome.

Supplemental Table 5. Diagnostic/procedure codes used to define outcomes.

Supplemental Table 6. Extent of missingness per variable.

Supplemental Table 7. Baseline variables included in multinomial logistic regression models for creation of generalized propensity scores.

Supplemental Table 8. Proportional hazards violation testing results.

Supplemental Table 9. Baseline characteristics stratified by UACR subgroup.

Supplemental Table 10. Unweighted and weighted event rates by study drug group and by KFRE subgroup separately.

Supplemental Table 11. Comparative effectiveness of insulin glargine (n=54,015) versus SGLT2i (reference group, n=84,625) and insulin glargine (n=54,015) versus non-exendin GLP-1 RA (reference group, n=21,788) in the entire cohort.

Supplemental Table 12. Comparative effectiveness and study drug-by-KFRE interaction P values for insulin glargine (n=54,015) versus SGLT2i (reference group, n=84,625) and insulin glargine (reference group, n=54,015) versus non-exendin GLP-1 RA (n=21,788) pairwise comparisons across KFRE subgroups.

Supplemental Table 13. Class switching over time among discontinuers of GLP-1 RA or SGLT2i.

Supplemental Table 14. Comparative effectiveness of SGLT2i (n=84,625) versus non-exendin GLP-1 RA (reference group, n=21,788) across eGFR subgroups.

Supplemental Table 15. Comparative effectiveness of SGLT2i (n=84,625) versus non-exendin GLP-1 RA (reference group, n=21,788) across UACR subgroup.

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